EAST WEST UNIVERSITY

Report Paper of Mini-Project

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Introduction:

Our project's objective is to explore the randomly generated undirected graph that is represented by an adjacency matrix that has been done using C program for a known vertex count. Our main objectives are to determine the number of edges and the degrees of all vertices in a graph, and show that the handshaking logic holds and access the computational time. Furthermore, we aim to show the computational time vs n in a graph and determine its time complexity.

Degrees and time complexity:

After carefully analyzing our code's output, we were able to determine the edges and degrees of all vertices in the graph for different values of n. We discovered that the degrees were double of the edges and that the handshaking logic holds for the different n values. We observed that the processing time increased as number of vertices (n) increased. Our highest time complexity observed was Big O (n^2), which we estimate to be time complexity.

Output of the Program:

For n= 1000 vertices

```
Enter the number of vertices (max 5000): 1000

Sum of degrees: 499766
Number of edges: 249883
The handshaking logic holds
Computational Time: 48.000000 ms
```

For n= 2000 vertices

```
Enter the number of vertices (max 5000): 2000

Sum of degrees: 1999202
Number of edges: 999601
The handshaking logic holds
Computational Time: 203.000000 ms
```

For n= 3000 vertices

```
Enter the number of vertices (max 5000): 3000

Sum of degrees: 4498852
Number of edges: 2249426
The handshaking logic holds
Computational Time: 442.000000 ms
```

For n= 4000 vertices

```
Enter the number of vertices (max 5000): 4000

Sum of degrees: 7998024
Number of edges: 3999012
The handshaking logic holds
Computational Time: 935.000000 ms
```

For n= 5000 vertices

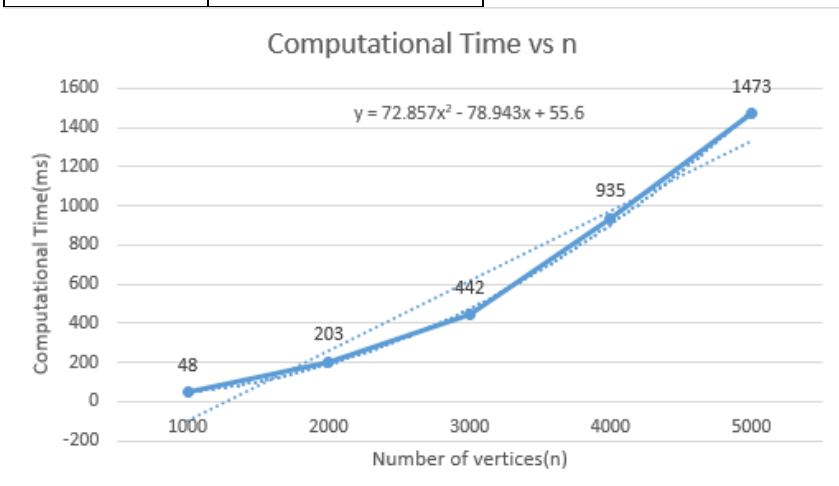
```
Enter the number of vertices (max 5000): 5000

Sum of degrees: 12497076
Number of edges: 6248538
The handshaking logic holds
Computational Time: 1473.000000 ms
```

Computational Time vs n Graph:

Number of vertices(n)	**Computational Time(ms)**
1000	**48**

2000	**203**
3000	**442**
4000	**935**
5000	**1473**



Graph Observation: This graph shows us the relationship between the computational time (ms) with the number of vertices (n). It shows us that the computational time increases as the number of vertices increases.

Conclusion: So, from the graph we get the complexity as a function of n, f(n) $=72.857n^2 - 78.943n + 55.6 = O(n^2)$, and theoretically we get the time complexity, $f(n)=O(n^2)$. So, both the time complexity is the same from the program.

Thank You